

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:BE01000081

Date:08-01-2025

Subject Name:Basic Mechanical Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Use of Steam Tables is permitted.

- Q.1** (a) Explain with neat sketch: Open, Close and Isolated system. 03
- (b) Find expression of work done and change in internal energy for adiabatic process. 04
- (c) Discuss the primary causes of global warming. What are impacts of global warming? 07

- Q.2** (a) Write the statements for Boyle's law, Charles's law and Gay-Lussac's law. 03
- (b) What do you mean by non metallic materials? Name any three non metallic materials and state its practical importance. 04
- (c) With a neat sketch explain Throttling calorimeter. 07

OR

- (c) 5 kg of air is heated from 25°C to 150°C . Determine: (i) specific heats, (ii) change in internal energy, (iii) change in enthalpy and (iv) heat supplied. Assume $R = 0.287 \text{ kJ/kgK}$ and $\gamma = 1.4$ for air and work done = 500 kJ. 07
- Q.3** (a) State the assumptions made in working of Carnot cycle. Also write down the limitations of Carnot cycle. 03
- (b) An engine operation on an air-standard Otto cycle has a compression ratio equal to 7. The conditions at the start of compression are 0.1 MPa and 300 K. The pressure at the end of heat addition is 4 MPa. Determine: (i) thermal efficiency, (ii) net work done per kg of air. Take $C_v = 0.718 \text{ kJ/kg}$, and $\gamma = 1.4$ 04
- (c) Derive equation for air standard efficiency of diesel cycle. 07

OR

- Q.3** (a) Give difference between SI and CI engines. 03
- (b) The following data were noted for a 4-cylinder, 4-stroke engine: 04
Diameter = 101 mm, stroke = 114 mm, engine speed = 1600 rpm, fuel consumption = 0.204 kg/min, heating value of fuel = 41,800 kJ/kg, difference in either side of the brake pulley = 378 N, brake pulley radius = 3.35 m. Assume mechanical efficiency = 83%.
Calculate: (i) brake thermal efficiency, (ii) indicated thermal efficiency, (iii) fuel consumption per brake power.
- (c) Explain with neat sketch four stroke diesel engine. 07

- Q.4 (a)** What is priming? Why is it required in centrifugal pump? **03**
(b) Explain with P-V diagram need of multi staging in reciprocating air compressor. **04**
(c) Explain with neat sketch working of Babcock and Wilcox boiler. **07**

OR

- Q.4 (a)** Define boiler. Also mention at least five essential requirements of a good boiler. **03**
(b) Give comparison between reciprocating and centrifugal pump. **04**
(c) What do you mean by mounting and accessories? Enlist different boiler mountings and accessories with their functions. **07**

- Q.5 (a)** Define the terms: Refrigeration, Refrigeration effect and Coefficient of Performance. **03**
(b) Write down comparison between Belt, Chain and Gear drive. **04**
(c) Explain with neat sketch vapour compression refrigeration system. **07**

OR

- Q.5 (a)** Define air conditioning. Also give the classification of air conditioning system. **03**
(b) Explain with a neat sketch working of internal expanding shoe brake. **04**
(c) What do you mean by the word 'Friction clutches'? Explain with neat sketch working of a centrifugal clutch. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000081

Date:09-06-2025

Subject Name:Basic Mechanical Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Discuss Closed, Open and Isolated Thermodynamic system with neat sketch.	03
	(b) Explain different between renewable and non-renewable energy.	04
	(c) Derive the characteristics gas equation for a perfect gas with usual notations.	07
Q.2	(a) State and explain zeroth law of thermodynamics and First law of thermodynamics.	03
	(b) With usual notations prove that $C_p - C_v = R$.	04
	(c) Write a difference between SI engine and CI engine.	07
	OR	
	(c) A sample of wet steam at pressure of 15 bar has a dryness fraction of 0.06. determine the enthalpy, volume, entropy and internal enthalpy per kg.	07
Q.3	(a) What is dryness fraction? Also state its importance.	03
	(b) List essential qualities of a good boiler.	04
	(c) One kg of air at 7 bar pressure and 90° C temperature undergoes a non-flow process. The law of expansion is $PV^{1.1} = C$. the pressure falls to 1.4 bar during process calculate (1) the final temperature (2) work done (3) Change in internal energy (4) Heat exchange. Take $R = 287 \text{ J/kg.K}$ and $\gamma = 1.4$ for air.	07
	OR	
Q.3	(a) What is priming? Why priming is required in centrifugal pump?	03
	(b) List different mountings of boiler and explain any one in brief.	04
	(c) With usual notation derive the expression for air standard efficiency of diesel cycle.	07
Q.4	(a) What is the difference between rigid coupling and flexible coupling?	03
	(b) Explain working of Vapor absorption refrigeration system with neat sketch.	04
	(c) One kg of an ideal gas is heated from 18° C to 98° C assuming $R = 0.27 \text{ KJ/kgK}$ and $\gamma = 1.18$ for gas calculate (1) specific heats (C_p and C_v) (2) Change in internal energy (3) change in enthalpy.	07
	OR	
Q.4	(a) Write the function of following I.C. Engine components: (1) Spark Plug (2) Injector (3) Piston Rings	03
	(b) Define the terms: Hardness, Toughness, Ductility, Elasticity	04
	(c) It is required to find out the efficiency of an air standard Carnot Cycle with the following data. Minimum temperature of cycle 15° C, minimum pressure in cycle 1 bar. Pressure after isothermal compression 3.5 bar. Pressure after isentropic compression 10.5 bar. Assume $R = 0.287 \text{ KJ/kgK}$ for air. What power would be produced if engine makes 2 cycles/second.	07

- Q.5** (a) What is throttling calorimeter? Also state its limitation **03**
(b) Explain working of split air conditioner. **04**
(c) Calculate the dimensions (diameter and stroke) of a double acting reciprocating compressor neglecting clearance volume from the following additional data: compressor power=10 kw, Compressor ratio=7, Inlet pressure=1 bar, Piston speed = 180 m/min, Law of compression = $PV^{1.3}=C$, $L/D = 1.5$ **07**

OR

- Q.5** (a) Define brake and also state difference between clutch and coupling. **03**
(b) Elaborate differences between reciprocating and rotary compressor? **04**
(c) A six-cylinder four stroke petrol engine develop 300 KW brake power at 2500 rpm. The stroke to bore ratio is 1.25. Assuming the mechanical efficiency as 80% and mean effective pressure of 9 bar, determine the bore and stroke of engine. Also find the fuel consumption in kg/hr if indicated thermal efficiency is 30% and CV of fuel used is 41900 KJ/kg. **07**

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000081/BE01R00081

Date: 29-12-2025

Subject Name: Basic Mechanical Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.
5. Use of Steam Tables is permitted.

	Marks
Q.1 (a) Classify the Babcock-Wilcox boiler. What is the function of a superheater?	03
(b) Describe advantages and disadvantages of solar energy.	04
(c) (i) A 3-kg plastic tank that has a volume of 0.2 m^3 is filled with liquid water. Determine the weight of the combined system. (ii) A pressure gage connected to a tank reads 500 kPa at a location where the atmospheric pressure is 94 kPa. Determine the absolute pressure in the tank. (iii) What is the characteristic of a steady-flow process?	07
Q.2 (a) State the function of any three accessories in boilers.	03
(b) What is the difference between elasticity and plasticity? What do you understand by resilience?	04
(c) Air enters a compressor at 1.013 bar and 27°C having a volume of $5 \text{ m}^3/\text{kg}$ and it is compressed to 12 bar isothermally. Determine: (i) work done, (ii) heat transfer, and (iii) change in internal energy.	07
OR	
(c) Five kg of air is heated from an initial volume of 0.5 m^3 to final volume of 1.3 m^3 at constant pressure 4 bar. Determine: (i) heat supplied, (ii) work done, and (iii) initial and final temperature of air. $C_p = 1.005 \text{ kJ/kg K}$, $R = 0.287 \text{ kJ/kg K}$.	07
Q.3 (a) Draw a schematic diagram of a differential band brake.	03
(b) Show the vapour compression refrigeration cycle on T-s and p-h diagram. Name each process in the cycle.	04
(c) Steam at 5 bar and dryness fraction of 0.85 expands reversibly at constant pressure until the temperature is 220°C . Determine the work and heat transfer per kg of steam during the process. Take specific heat for superheated steam as 2.1 kJ/kg K .	07

OR

- (a) Answer in one word: 03
- (i) Which coupling is used to connect engine shaft to driven shaft in automobile?
- (ii) Energy absorbed by a brake is released into the environment in the form of which energy?
- (iii) What causes the outward movement of a shoe in centrifugal clutch?

- (b) A classroom has a window air conditioner of 2 TR capacity. Its COP is 3. How much electrical energy (in kWh) will be consumed in one month if the unit runs for 6 hours per day? 04

- (c) During a test of steam with throttling calorimeter, sample is taken from main pipe at 15 bar, pressure and temperature after throttling are 1.2 bar and 150°C respectively. Determine dryness fraction of steam. Take specific heat of superheated steam as 2.1 kJ/kg K. 07

- Q.4** (a) What is a flexible coupling? What are its types? 03

- (b) Draw a diagram of vortex-type centrifugal pump. 04

- (c) A steam plant works on ideal Rankine cycle. Steam is supplied at 25 bar and 290°C. The steam exhausts into a condenser at 15 kN/m² with 0.9 dry. Determine the cycle efficiency. 07

OR

- (a) Give similarities and differences between a friction clutch and a positive-contact clutch. 03

- (b) Describe in short the parts of a reciprocating pump. 04

- (c) An oil engine working on diesel cycle has cylinder bore of 190 mm and piston stroke of 230 mm. The clearance volume is 290 cm³. The fuel injection takes place at a constant pressure for 6% of the stroke. Determine the air-standard efficiency. 07

- Q.5** (a) Answer in short: 03

(i) Name the different types of belt drives.

(ii) Enlist different gear types.

(iii) Give one benefit of using a V-belt.

- (b) What do you understand by positive displacement compressor? How it differs from rotodynamic compressor? 04

- (c) Explain following terms of I.C. engine: (i) brake power, (ii) indicated power, (iii) specific fuel consumption, (iv) clearance volume. 07

OR

- (a) State one disadvantage of rope drive, chain drive and gear drive. 03

- (b) Define with regard to compressors: double-acting compressor, multistage compressor, displacement (swept) volume, pressure ratio. 04

- (c) A 6-cylinder 4-stroke I.C. engine is used to produce 95 kW brake power at 800 rpm. The stroke to bore ratio is 1.25. Mean effective pressure is 7 bar. Determine the bore and stroke of the engine. Assume mechanical efficiency as 80%. 07
